

Multimodal Feature Fusion in Mixed Reality for Detection of Neurodegenerative Disorders

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ABSTRACT

Neurodegenerative disorders such as Parkinson's disease involve progressive motor, speech, and balance impairments, whose heterogeneity complicates but underscores the importance of early diagnosis and monitoring. In this study, we present a Mixed Reality (MR)-based framework that leverages Microsoft HoloLens 2 for multimodal, real-time acquisition of behavioral and physiological signals during ecologically valid tasks. We conducted a study involving 137 participants (66 diagnosed with Parkinson's and 71 healthy controls), where MR was employed for both data collection and task presentation. After preprocessing and feature extraction, we applied machine learning models combining deep learning and gradient boosting methods, alongside early- and late-fusion strategies, to enhance classification performance. Our results indicate that multimodal fusion substantially outperforms single-modality approaches, with early fusion reaching high performance on both accuracy and F1-score, exceeding 96%. Moreover, MR-based acquisition proved alternative to conventional smartphone- and smartwatch-based systems by enabling synchronized, multidimensional, real-time data capture across multiple modalities.

Index Terms: virtual reality, augmented reality, Parkinson's disease, feature engineering, multidimensional analysis, signal processing

1 INTRODUCTION

Parkinson's disease (PD) produces characteristic motor and speech changes that can be quantified from movement and language signals. Wearable- and smartphone-based systems have already demonstrated high sensitivity and specificity [22, 16, 20]. However, translating these approaches into routine use remains difficult. Patient burden can reduce adherence, and clinician burden limits uptake [27, 18, 21]. In this context, Mixed Reality (MR) offer a complementary path toward unobtrusive, repeatable assessment with minimal setup, suitable for scalable screening. Yet prior work commonly focuses on unimodal pipelines or evaluates a single fusion pathway, making it difficult to compare methods fairly or understand which modalities and fusion choices generalize best. Here, we deliver a unified, reproducible study that (i) quantifies univariate separability across thousands of features, (ii) compares multiple fusion strategies under stratified cross-validation, and (iii) calibrates operating thresholds to maximize F1 in screening-oriented use cases.

2 RELATED WORK

The pursuit of objective biomarkers for neurodegenerative disorders (ND) has catalyzed innovation at the intersection of clinical neuroscience, sensor technology, and computational science. The current state of the art is characterized by three parallel

streams of development: the use of wearable sensors for objective symptom quantification, the application of immersive realities to create valid assessment paradigms, and the deployment of advanced machine learning techniques for data analysis and classification. This section provides a review of each of these domains, establishing the scientific and technical foundation upon which the present work is built.

2.1 Objective Symptom Quantification with Wearable Sensors

The widespread availability of micro-electro-mechanical systems (MEMS) has driven the adoption of body-worn sensors as a primary tool for objective motor assessment in NDs. These systems, which are typically built around with the use of accelerometers and gyroscopes, are embedded in a variety of form factors, from consumer devices like smartphones and smartwatches to specialized, medical-grade monitors such as the PKG watch or the multi-sensor PDMonitor system. Mentioned in the articles medical-grade monitors like Kinesia 360 has the Area Under the Curve (AUC) for tremor detection: 0.89. AUC for dyskinesia: 0.86. AUC for bradykinesia: 0.82. The PDMonitor consisted of 5 sensors achieve Machine Learning (ML) accuracies: Tremor (0.87), Dyskinesia (0.85), Bradykinesia (0.745), and Freezing of Gait (0.96). [21, 6].

Deep learning systems using wearable sensors demonstrate variable but generally high classification performance for Parkinson's disease detection, with performance ranging from 65% to 98% accuracy depending on classification task complexity. Binary classification of PD versus healthy controls achieves the highest performance, with AUCs ranging from 0.888 to 0.998, while multi-class motor state detection shows moderate performance (0.654 balanced accuracy for three-class classification) and specific symptom detection like freezing of gait achieves 90% geometric mean of sensitivity and specificity [21, 6, 9, 17, 19, 25].

Wearable devices effectively differentiate PD patients from healthy controls with high sensitivity and specificity. These devices enable the precise, quantitative measurement of the cardinal motor symptoms of PD, including tremor frequency and amplitude, bradykinesia (e.g., through finger tapping speed), and a host of gait disturbances, such as reduced stride length, increased gait variability, and postural instability [22, 16, 20].

Despite their accuracy, translating these tools to routine clinical practice faces significant hurdles as patient factors (discomfort and the social stigma of visible medical technology often lead to poor compliance) and clinical factors (high-frequency data streams can overwhelm clinicians, who may lack the specialized tools or time to integrate this information into decision-making workflows) [27, 18, 21].

This creates an "objectivity paradox": the technology's promise of unbiased data is often overshadowed by the subjective experiences of everyday use. For patients, the device can be a source of emotional distress and a constant reminder of their condition. Furthermore, the sensors used in research are often attached to various parts of the body, forcing patients to disrobe and causing additional discomfort. Therefore, successful tech-



Figure 1: Women in the healthy control group (63 years old), taking the test with AR.

nology must not only be objectively accurate but also designed to integrate seamlessly into a patient's life and build trust with clinicians [18, 21].

2.2 Immersive Realities for Valid Assessment

Immersive technologies—Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR)—bridge the gap between laboratory control and real-world functional relevance. By simulating complex daily activities like obstacle navigation or cognitive dual-tasking in a safe, reproducible digital space, these platforms provide more controlled assessment environments than traditional clinical settings [23, 12, 2, 11]. In neurology, VR is utilized for cognitive and post-stroke rehabilitation, while AR/MR tools address motor impairments in Parkinson's Disease (PD), such as freezing of gait (FOG) [12, 23, 11, 10].

The Microsoft HoloLens serves as a high-precision platform for clinical PD assessment, utilizing integrated head, eye, and hand-tracking sensors to quantify performance on standard tests like the Timed Up and Go (TUG) and Five Times Sit To Stand (FTSTS) [28, 4]. Custom algorithms applied to HoloLens data can extract biomechanical features from tasks like finger tapping, postural tremor, and eye-hand coordination, demonstrating concurrent validity with gold-standard motion capture systems [7, 13].

Clinical scales and wearables are often affected by environmental variability or a lack of context. However, an MR system functions as a "standardized context engine." By projecting identical holographic stimuli and interactive environments for each participant, MR isolates patient performance as the primary variable. This standardization allows for more sensitive and reliable longitudinal tracking of disease progression and therapeutic response. Furthermore, the MR system incorporated in the HoloLens requires only that it be placed on the head; all necessary sensors are built into the device, which limits the discomfort caused by the amount or placement of wearable sensors. [18, 7].

Authors in [3] introduces the use of wearable Mixed Reality (MR) for cognitive training and potential diagnostic monitoring in patients with early-stage Alzheimer's Disease (AD) - MemHolo. Based on the article, the user experience (UX) of the HoloLens for elderly patients with neurodegenerative conditions is characterized by high levels of curiosity and engagement, though it presents specific ergonomic and interaction challenges. Patients generally perceived the HoloLens as a "futuristic" and "entertaining" device, with many expressing pride in participating in scientific research using technology their grandchildren had never used. Unlike traditional Virtual Reality (VR) systems, participants reported no motion sickness or digital eye strain during MemHolo sessions. This is attributed to the "Mixed Reality" nature of the device, which allows users to remain aware of their physical surroundings and caregivers, reducing the "isolation effect" common in immersive VR.

3 MR SYSTEM DATA

3.1 Acquisition

Speech and behavioral signals were acquired using a Microsoft HoloLens 2 running a custom Unity application. The MR headset presented audiovisual task instructions while concurrently logging three synchronized, time-stamped streams.

Throughout the session, the HoloLens 2's suite of sensors captures multimodal data during 17 different tasks: movement and orientation via the IMU (accelerometer/gyroscope), hand and body motion via depth and RGB cameras, eye movements via infrared sensors, and voice via microphones.

- **Audio:** Captured via the integrated five-element directional microphone array. Tasks: Description of the picture (Describe-image); Question about the examinee's day (Describe-day); Repetition of the memorized text (Describe-story); Vowels (, a, e, i, o, u); Repetition of PETAKA words (pakata, pataka, petaka)
- **Kinematics:** Head pose and hand-motion acceleration from onboard inertial sensors. Tasks: Touching the first and second fingers, right and left hand (Tapping); Opening the palm, right and left hand (Hand-closing, Diadochokinesis); Light bulb, right and left hand (Hand-rotations, Diadochokinesis); Holding hands in front of the body (Postural-tremor); Exercise with armrests from a chair (Rest-tremor); Touching the nose with a finger (Touching-nose); Walking 12 m, stopping when standing up (Gait); Walking 3 m, without stopping (Gait)
- **Eye-Tracking:** Gaze points and movement traces from the built-in tracking module. Tasks: Tracking static points (REFLEX); Opposite tracking of static points (ANTI); Tracking previous static points (MEMORY); Tracking dynamic points (MOVING)

While all data were sample-aligned for analysis, features derived from saccadic movements were excluded from this study due to limited recording quantity.

3.2 Patient data

The study recruited 66 patients with Parkinson's disease (PD; 53.50 ± 7.48 years) and 71 healthy controls (HC; 48.50 ± 9.08 years). Data were collected using mixed reality (MR) for task presentation. All PD participants were evaluated in the ON-medication state. To ensure tight clinical-behavioral correspondence, motor and cognitive assessments were conducted immediately prior to speech recordings. The clinical profile of the PD group (detailed in Table 1) was characterized by the neurologist using scales:

- **Motor Impairment (MDS-UPDRS-III):** Participants exhibited mild ($n = 37$; scores 0–32) or moderate ($n = 29$; scores 33–58) impairment; no severe cases (> 58) were present.
- **Cognitive Function (MoCA):** Status ranged from normal ($n = 11$; scores 26–30) to mild ($n = 30$; 18–25) and moderate ($n = 25$; 10–17) impairment; no severe impairment (< 10) was observed.
- **Voice Assessment:** The Voice Handicap Index (VHI) and the Voice Disability Coping Questionnaire (PL-VDCQ) were used to evaluate the psychosocial impact and coping strategies related to voice disorders.

Healthy controls reported no neurological diagnoses or functional limitations. In [14] authors describe the user experience of XR diagnostic platform for Parkinson's Disease (PD) implemented on Microsoft HoloLens 2 used in this study. By integrating visual, auditory, and textual guidance, the system enhances user comprehension and reduces the anxiety typically associated with clinical assessments, fostering a sense of autonomy and empowerment among patients. Results from a study involving 15

PD patients and 32 healthy controls indicate high engagement and reduced cognitive load through simplified interactions like button presses.

Table 1: Clinical assessment scores for the Parkinson's disease group (N = 66).

Scale	Mean	Std. Deviation
MDS-UPDRS-III	28.75	11.07
VHI	11.68	4.75
PL-VDCQ	15.05	4.21
MoCA	20.24	3.82

4 FEATURE EXTRACTION AND FUSION STRATEGIES

We integrated curated parameter tables for PD and HC ($N = 137$), retaining only numeric features. To ensure data quality, we excluded participants and features with $> 50\%$ missing values. Remaining missingness was addressed using median imputation for statistical summaries and a Histogram Gradient Boosting regressor for the DNN pipeline to provide more reliable, column-wise estimates.

The resulting design matrix comprised $\sim 4.8k$ features across four primary modalities:

- **Head-IMU:** Median, RMS, and jerk statistics for acceleration and angular velocity (e.g., `HEAD_median_acc`, `HEAD_rms_gyro`).
- **Speech & Linguistics:** Prosodic markers (speech rate, silence ratio, fillers) and linguistic complexity metrics (Type-Token Ratio (TTR), Measure of Textual Lexical Diversity (MTLD), and Part-of-Speech (POS) distributions).
- **Tremor & Gait:** Features derived from postural/resting tremor, finger tapping, and tactile tasks.
- **Task-Based Context:** Features were extracted across multiple tasks, including sustained vowels, Diadochokinesis (DDK), resting tremor assessment, and narrative tasks (Describe-day, Describe-image)

We evaluated several fusion strategies to combine these high-dimensional blocks.

- **Block-Scaled Early Fusion-** per-modality z -scoring, followed by block scaling $1/\sqrt{p_k}$ (where p_k is the block dimensionality), and then L2 logistic regression. This limits the dominance of high-dimensional blocks.
- **Intermediate Fusion** employed Canonical Correlation Analysis (CCA) to project the strongest modalities into a $d = 2$ latent space. The two strongest modality-views (by median univariate AUC) are standardized and projected via two-view CCA ($d=2$ components per view); classification is performed on $[U||V]$ using L2 logistic regression. This provides interpretable latent spaces.
- **AUC-Weighted Late Fusion** - per-modality L2 logisticians are trained; within each CV fold, test-fold probabilities are averaged using weights proportional to the training-fold AUC of each modality (no meta-learner).

We utilized a ResNet-based architecture to extract embeddings from normalized feature matrices. These embeddings were either classified independently or concatenated for multi-modality fusion. Decision thresholds (t) were optimized on out-of-fold Precision-Recall curves to maximize F_1 scores.

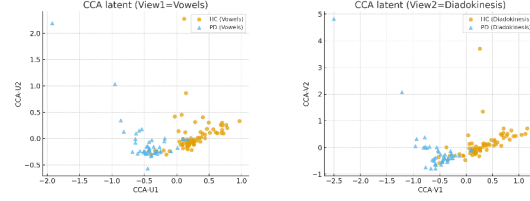


Figure 2: CCA latent spaces for two strongest views (Vowels and Diadochokinesis); clear PD/HC contrast in canonical variates.

5 RESULTS

5.1 Single features DNN classification

Analysis of Table 2 reveals that kinematic features outperformed speech-based models. The Head feature group yielded the highest scores reaching an F_1 -score of 97.8% and up to 98.0% accuracy. The tremor feature group also demonstrated strong diagnostic performance, achieving F-scores ranging from 91% to 95%, with postural tremor reaching 100% precision and hand tremor achieving 100% recall. Conversely, the Speech group underperformed. Vowel tasks showed the highest variability, with F_1 -scores ranging from 59.3% (vowel ‘e’) to 71.4% (vowel ‘i’). However, naturalistic speech tasks were more robust: “describe-image” reached 94.4% F_1 and 100% recall, whereas “describe-day” achieved 100% precision with a 91.7% F_1 -score.

Table 2: Performance of individual features (F1-score, accuracy, precision, recall).

	Features	Best F1	Accuracy	Precision	Recall
Speech	a	0.693 ± 0.106	0.984 ± 0.031	1 ± 0	0.964 ± 0.071
	e	0.593 ± 0.231	0.680 ± 0.164	0.698 ± 0.316	0.570 ± 0.273
	i	0.714 ± 0.095	0.750 ± 0.129	0.793 ± 0.239	0.700 ± 0.135
	o	0.694 ± 0.061	0.625 ± 0.088	0.534 ± 0.069	1 ± 0
	u	0.867 ± 0.163	0.97 ± 0.096	0.917 ± 0.167	0.875 ± 0.250
	Describe-day	0.917 ± 0.167	0.938 ± 0.125	1 ± 0	0.875 ± 0.25
	Describe-image	0.944 ± 0.064	0.950 ± 0.058	0.900 ± 0.115	1 ± 0
	Describe-story	0.917 ± 0.061	0.921 ± 0.060	0.950 ± 0.100	0.902 ± 0.122
	pakata	0.781 ± 0.081	0.833 ± 0.010	0.940 ± 0.125	0.708 ± 0.210
	pataka	0.811 ± 0.070	0.820 ± 0.058	0.767 ± 0.155	0.900 ± 0.140
Tremor	petaka	0.747 ± 0.096	0.781 ± 0.120	0.875 ± 0.250	0.729 ± 0.208
	Hand	0.950 ± 0.100	0.969 ± 0.062	0.917 ± 0.0167	1.00 ± 0
	Postural	0.930 ± 0.045	0.936 ± 0.042	1 ± 0	0.875 ± 0.083
	Rest-tremor	0.949 ± 0.058	0.95 ± 0.058	0.958 ± 0.083	0.950 ± 0.100
Gait	Touching	0.910 ± 0.064	0.911 ± 0.068	0.873 ± 0.093	0.964 ± 0.071
	gait	0.860 ± 0.104	0.875 ± 0.122	0.875 ± 0.160	0.873 ± 0.089
Head	HEAD	0.978 ± 0.029	0.980 ± 0.023	0.980 ± 0.023	0.964 ± 0.042

5.2 Fusion strategies

We evaluated three strategies to address high-dimensional multimodal imbalances. AUC-Weighted Late Fusion achieved the highest overall performance ($F_1 = 0.950$, Recall= 1.000), demonstrating that ensembling per-modality classifiers via a discrimination-driven linear pool effectively harnesses complementarity without overfitting. Block-Scaled Early Fusion provided a robust single-model baseline (ROC-AUC= 0.974, $F_1 = 0.910$), successfully equalizing the influence of high-cardinality feature blocks. While CCA-Based Intermediate Fusion yielded more modest predictive gains (ROC-AUC= 0.915), it produced highly interpretable latent spaces that clearly separated PD from HC (see Fig. 2, Table 3). All methods use 5-fold stratified cross-validation; metrics are computed on out-of-fold (OOF) under subject-wise split predictions.

5.3 Multi-features DNN classification

Based on results in Table 2, data fusion was prioritized to bolster lower-performing speech predictors by integrating them with Postural, Touching, and Gait modalities to capture complementary physiological signals. As shown in Table 4, fusing speech with the Postural task consistently yielded F_1 -scores above 90%, with the “a” fusion achieving a 28.3% improvement over its unimodal baseline. Significant synergistic gains were observed when combining diadochokinesis (DDK) with kinematic descriptors. The fusion of petaka with postural features achieved

Table 3: Fusion strategies (5-fold OOF) on all features. BS-EF – Block-scaled Early Fusion; CCA-Int – Canonical Correlation Analysis Intermediate Fusion (Vowels + Diad.); AUC-LF – AUC-weighted Late Fusion.

Metric	BS-EF	CCA-Int	AUC-LF
Acc.	0.912 ± 0.010	0.788 ± 0.051	0.949 ± 0.032
F_1	0.910 ± 0.005	0.752 ± 0.012	0.950 ± 0.010
Prec.	0.897 ± 0.008	0.863 ± 0.034	0.904 ± 0.021
Rec.	0.924 ± 0.013	0.667 ± 0.025	1.000 ± 0.000
ROC–AUC	0.974 ± 0.011	0.915 ± 0.015	0.968 ± 0.009

an F_1 -score of 96%, while pataka combined with postural features and pakata combined with gait features both exceeded 90% F_1 , outperforming their respective single-modality models. While isolated vowels generally remained resistant to substantial gains, the fusion of the vowel "a" with Postural reached an F_1 of 97.7%, surpassing both standalone modalities. These findings demonstrate that multimodal integration—specifically pairing naturalistic speech or rapid syllables with motor-kinematic data—effectively mitigates individual biomarker variability to provide a more robust diagnostic profile.

Table 4: Performance of speech features combined with features related to other functionalities, measured using the F_1 -score. Values in the table are the average across all fusions. Values exceeding the performance of all individual feature models are highlighted in green. Values surpassing at least one of the individual feature models in terms of F_1 -score are highlighted in blue.

Speech features	Single	Postural	Gait	Touching
a	0.693 ± 0.106	0.977 ± 0.045	0.864 ± 0.048	0.912 ± 0.068
e	0.574 ± 0.231	0.921 ± 0.053	0.882 ± 0.140	0.829 ± 0.182
i	0.714 ± 0.095	0.933 ± 0.097	0.853 ± 0.139	0.900 ± 0.115
o	0.701 ± 0.068	0.944 ± 0.064	0.922 ± 0.097	0.891 ± 0.084
u	0.867 ± 0.163	0.839 ± 0.236	0.914 ± 0.102	0.881 ± 0.158
Describe-day	0.917 ± 0.167	0.958 ± 0.049	0.86 ± 0.082	0.939 ± 0.079
Describe-image	0.944 ± 0.064	0.954 ± 0.052	0.923 ± 0.109	0.917 ± 0.061
Describe-story	0.917 ± 0.061	0.958 ± 0.049	0.901 ± 0.107	0.894 ± 0.079
pakata	0.780 ± 0.081	0.929 ± 0.082	0.902 ± 0.122	0.874 ± 0.146
pataka	0.811 ± 0.070	0.922 ± 0.097	0.867 ± 0.163	0.850 ± 0.117
petaka	0.747 ± 0.096	0.977 ± 0.045	0.854 ± 0.118	0.894 ± 0.082

6 DISCUSSION

Multimodal fusion with MR-based acquisition consistently improved discrimination over single-modality baselines under our subject-wise, stratified evaluation. Across classical ML models, block-scaled early fusion mitigated the dominance of high-cardinality blocks and delivered strong performance (ROC–AUC $\approx$ 0.97, F_1 =0.910; Table 3), while an AUC-weighted late-fusion ensemble capitalized on complementary views to achieve the highest F_1 with perfect recall in our data (F_1 =0.950, recall = 1.000; Table 3). In practice, early fusion is a robust default; block scaling and/or AUC-weighted late fusion can be added to improve operating characteristics. Threshold tuning should be pre-specified (OOF-based) to target F_1 or recall.

Feature extraction and classification using a pretrained ResNet yielded very strong results. Within each feature group—speech, tremor, gait, and head—there was at least one feature on which classification achieved a minimum of 90% for both F_1 -score and accuracy. Furthermore, the fusion of ResNet-extracted features demonstrates high effectiveness. In many cases, fusion improved at least one performance metric, which can already be considered a substantial success. In several tests, fusion even produced higher F_1 -scores than those obtained from the single-feature classifications of both modalities. These patterns align with prior evidence that wearable inertial and kinematic signals are highly informative for PD motor manifestations [21, 6, 27, 16, 29, 1, 22, 20, 18, 5] and that multimodal

models outperform unimodal counterparts by capturing complementary physiology [8, 24, 15, 26].

Finally, MR-based acquisition standardizes stimuli and context while synchronizing multi-sensor capture [23, 12, 2, 11], a potential advantage over phone/watch pipelines, which is a potential advantage over phone/watch pipelines. Another advantage is comparable classification results, which are mostly improved, compared to systems based on wearable sensors [21, 6, 27, 9, 17, 19, 25]. Adding user experience opinions, such as those in [3, 14] which state high levels of curiosity and engagement, enhanced user comprehension, and reduced anxiety typically associated with clinical assessments, we foster a sense of autonomy and empowerment among patients. This makes MR-based acquisition an attractive alternative to classical examinations and wearable sensor-based approaches.

Next, we will validate generalization in prospective multi-site studies across languages and HMDs, expanding cohorts to prodromal/severe PD, atypical parkinsonism, ON/OFF levodopa challenges, and neurological controls (e.g., essential tremor), with rater-adjudicated and continuous UPDRS labels. We will compare tabular vs temporal models and self-/contrastive objectives across modalities.

Future evaluation will prioritize calibrated, decision-ready outputs and head-to-head cost/effort comparisons.

7 CONCLUSION

This study introduces a novel framework for the multimodal assessment of neurodegenerative disorders, demonstrating the exceptional capability of a Mixed Reality-based system. Our findings show that multimodal feature fusion significantly outperforms single-modality approaches, with early fusion reaching high performance on both accuracy and F_1 -score, exceeding 96%. This validates our core hypothesis that a comprehensive, multi-sensor approach is crucial for capturing the complex, multi-system nature of conditions like Parkinson's disease.

The results further highlight the advantages of our MR-based data acquisition. By enabling synchronized, multidimensional, real-time data capture across various modalities, our system provides a richer and more robust data stream, which is a critical advancement for developing more sensitive and reliable diagnostic tools. Specifically, modalities related to head dynamics and gait proved to be powerful individual predictors, with F_1 -scores reaching 96% and 92%, respectively. While speech features showed lower individual performance, their fusion with other data streams demonstrated promising gains, with the "describe-day" feature reaching an F_1 -score of 93% and "describe-story" reaching 90%, reinforcing the value of combining multiple complementary data sources.

The article confirms the vast potential of MR-based systems for PD detection, which can outperform other methods in terms of achievable accuracy and other scores. MR-based systems evoke positive emotions and attitudes in patients toward such examinations. This can result in better cooperation between patients and doctors, as well as improved patient comfort and care as patients are not exposed to tedious and lengthy examinations or stigmatization.

In summary, this work provides a reproducible, unified framework that not only demonstrates the diagnostic potential of multimodal fusion in MR environments but also sets a new standard for data acquisition in neurodegenerative research. Our approach offers a path toward scalable, objective, and clinically meaningful screening tools that can seamlessly integrate into a patient's life while providing clinicians with transparent, actionable insights into disease progression.

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